CHECKLIST OF BBM104 SPRING 2025 PROJECT ASSIGNMENT 4

Please try the following test cases and drop your notes under each bullet point in **BLUE** color. You can see some examples below:

- Check that functionality x works.
  This functionality works like a charm.
- Check that functionality y works.
  This functionality has some flaws, the system crashes when it is tried to do operation namely t.
- Check that functionality z works.
  This functionality does not work at all.

You can add extra test cases please insert them as in **RED** color, your notes are still in **BLUE**. This paper is also for noting which parts of your system does not work well, so you can create hypothetic test cases that make your system not respond as it has to and explain the problem.

Test Cases

- **Contents of the assets folder must be same as the one provided at Piazza.**

The contents of the assets folder match the provided assets on Piazza.

Result: This functionality works as expected.

- **Open the terminal at your main folder (the folder that your code needs to be compiled and run)**

Terminal is opened at the main project folder where the code is compiled and run.

Result: This functionality works as expected.

- **Run the following commands:**
  - **javac -version (must be resulted same as "javac 1.8.0_xxx")**
  - **java -version (must be resulted same as "java version "1.8.0_xxx"")**

The following commands are executed:

```
PS C:\Users\ASUS\Desktop\src> javac -version
javac 1.8.0_431
PS C:\Users\ASUS\Desktop\src> java -version
java version "1.8.0_431"
```

Result: This functionality works as expected.

First and Last Name : Emre Türksoy
Student ID : 2230356030
Link of the Demo Video : https://youtu.be/izB2dJV2_Uw

CHECKLIST OF BBM104 SPRING 2025 PROJECT ASSIGNMENT 4

- **Compile your code and then run it.**

The code is compiled using "javac Main.java" and executed using "java Main".

Result: This functionality works as expected.

- **Show that the game opens without any crash. Your system must also read the assets by itself.**

The game launches without crashes, and the system automatically loads assets (e.g., tank images, bullet image, explosion images, wall image) from the assets folder with error handling.

Result: This functionality works perfectly. The game launches smoothly and loads all assets correctly.

- **Show that Player movement and shooting works correctly. The player tank should be correctly controlled with the movement buttons and it should fire correctly with the "x" button. Firing should not look erratic(no shooting rapid fire with no wait time). Tank movement should be accompanied by a simple animation of the tank using the images provided.**

Player movement (using arrow keys) and shooting (using the X key) are tested. The tank moves in the direction of the last pressed key, and shooting fires a bullet in the tank's current direction.

Result: This functionality works perfectly. Movement and shooting are smooth. A key list (pressedKeys) prioritizes the last pressed direction, and delta time ensures consistent movement and bullet speed across different frame rates.

- **Show that Enemy movement and shooting works correctly. Enemies should move randomly or semi randomly and they should shoot at semi random intervals. Keep in mind that tanks should not move too erratically(constantly changing directions multiple times in a single second) and they should not be able to fire again right after firing. Enemies should spawn correctly.**

Enemies move randomly every second and shoot at random intervals. Their movement and shooting behavior are tested for consistency. Also enemies spawn continuously every 3-8 seconds up to a maximum of 10 enemies.

```
// handle enemy spawning
if (enemies.size() < enemyCount && now - lastSpawnTime > (3_000_000_000L + Math.abs(rand.nextLong() % 5_000_000_000L))) { // 3-8 seconds
```

Result: This functionality works perfectly. Enemies move and shoot at correct intervals without glitches. Delta time ensures consistent movement and bullet speed.

First and Last Name : Emre Türksoy
Student ID : 2230356030
Link of the Demo Video : https://youtu.be/izB2dJV2_Uw

CHECKLIST OF BBM104 SPRING 2025 PROJECT ASSIGNMENT 4

- **Show that bullets behave correctly. Bullets should hit and destroy the correct type of tanks depending on who fires them. Bullet hit should be accompanied by appropriate explosion image.**

Bullets are tested for correct movement, collision with walls, and interaction with tanks (player and enemies). Upon collision, bullets trigger appropriate explosions and remove the target tank if applicable.

Result: This functionality works perfectly. Bullets move correctly, collide with walls and tanks, and trigger explosions (small for walls, large for tanks) as expected.

- **Show that walls are implemented. Walls should prevent tanks from passing through and cause bullets to disappear. There should be walls surrounding the map and inside the area itself.(You can place them as you wish)**

Walls are tested to ensure they block tank movement and destroy bullets upon contact.

Result: This functionality works perfectly. Walls are implemented correctly, preventing tanks from passing through and destroying bullets with a small explosion effect.

- **Scoring works correctly and lives decrease as necessary.**

Scoring increases by 10 points when an enemy is destroyed (via bullet or collision). Lives decrease when the player is hit by an enemy bullet or collides with an enemy tank.

Result: This functionality works perfectly. Scoring and lives update correctly based on game events.

- **Pause menu works correctly. Game can be resumed or closed or restarted**

The pause menu (triggered by pressing P) displays options to resume (P), restart (R), or close the game (Escape). All options are tested.

Result: This functionality works perfectly. The pause menu, resume, restart, and close functions operate without issues.

- **Game Over triggers correctly and Game Over Menu works correctly. Game can be closed or restarted.**

The game over screen triggers when lives reach 0, displaying the final score and options to restart (R) or close (Escape). Both options are tested.

Result: This functionality works perfectly. The game over screen triggers correctly, and restart/close options function as expected.

First and Last Name : Emre Türksoy
Student ID : 2230356030
Link of the Demo Video : https://youtu.be/izB2dJV2_Uw

CHECKLIST OF BBM104 SPRING 2025 PROJECT ASSIGNMENT 4

- **EXTRA TEST CASE: Vertical and Horizontal Scrolling with Camera Smoothing**

The game supports vertical and horizontal scrolling. The camera follows the player using linear interpolation (LERP) with a smoothing factor of 0.03, ensuring fluid visual transitions.

Result: This functionality works perfectly. The camera tracks the player smoothly across the map, enhancing the gameplay experience.

- **Moreover, please narrate the critical parts of your code (design), please do not narrate all of it, just narrate it for 1-2 minutes as it is just for understanding the code by your own perspective.**

Written Narration:

Game Loop (startGameLoop): The game loop uses AnimationTimer to handle updates. *Delta time ((now - lastUpdate) / 1_000_000_000.0)* ensures consistent movement and bullet speeds across different frame rates. Key mechanics like player-enemy collisions, bullet handling, and continuous enemy spawning are implemented here.

Player and Enemy Logic (Tank Class): The Tank class manages movement using a pressedKeys list to prioritize the last pressed direction. Enemies move randomly every second (now - lastEnemyMove > 1_000_000_000), and their shooting interval is randomized (0.4-0.7 seconds).

Camera Smoothing (Camera Class): The Camera class implements LERP (offset += (target - offset) * LERP_FACTOR) to smoothly follow the player, with a LERP_FACTOR of 0.03 for fluid scrolling.

Collision and Respawn: Player-enemy collisions reduce lives, trigger explosions, and initiate a 0.7-second respawn period (now - respawnStartTime >= 700_000_000L). During respawn, the player is immune to collisions.

Continuous Spawning: Enemies spawn every 3-8 seconds if the enemy count is below 10, using a randomized interval 3_000_000_000L + Math.abs(rand.nextInt() % 5_000_000_000L)).

Video Narration: In the gameplay video, I will explain these critical parts while demonstrating their effects in-game (e.g., showing smooth camera movement, enemy spawning, and collision effects).

First and Last Name : Emre Türksoy
Student ID : 2230356030
Link of the Demo Video : https://youtu.be/izB2dJV2_Uw

CHECKLIST OF BBM104 SPRING 2025 PROJECT ASSIGNMENT 4

INCOMPLETE PARTS

- --

First and Last Name : Emre Türksoy
Student ID : 2230356030
Link of the Demo Video : https://youtu.be/izB2dJV2_Uw

CHECKLIST OF BBM104 SPRING 2025 PROJECT ASSIGNMENT 4

RESTRICTIONS

- Your demo **must contain sound**, you can use Zoom's video recorder for this purpose, be sure about that you are sharing your **whole screen, not just the application.**
- Your demo video **cannot be longer than five minutes, four minutes for the GUI and one minute for code,** (3-4 minutes for the game and 1-2 minutes for the code is also OK but the total length of your video cannot be longer than five minutes) it cannot exceed five minutes even a little second, and your video must be in real speed, please do not speed up or down your videos.
- Everything about your **demo video must be in English** (you do not have to set your system language to English for sure). Your video must be fluent. It is not assumed that you are a native speaker, but it is known that your English level is at least B1+ (according to prep class), so, it is requested from you to speak at least as a B1+ speaker, little mistakes about your speech are going to be discarded but avoid mistakes as much as you can.
- **Do not do these operations too fast or too slow,** make them at the speed that a normal user **(who is not very familiar with your system and tries to read texts)** does.
- **Show the things that make your system crash or things that you couldn't done in demo properly** but show them after the requirements of the demo finishes, you can skip some of the requirements by saying that you did not implement it, so that your demo video becomes more fluid, less erroneous, but note that you must show your erroneous parts at the end.
- You must both narrate your GUI and code.
- **It is your own responsibility to show all the functionality of your system,** so if you have not mentioned about a functionality that is a request of this project or part of your extra feature, your code may be graded as low as zero for that part as there is no working example. So, please do not feel limited with given conditions, show your all work to get full credit, that means, if you have not shown a concept at your demo video, you will not have any right to say that **"But it worked on my machine!",** because this demo video is a chance for you to show that your code works on your machine well, but note that you can still have right to objection for the parts that are not mentioned at PDF, Piazza Notes, Q&As; say that there is a bug at a functionality and it is neither mentioned at PDF, Piazza, Q&A, nor mentioned during your implementation and you did not notice that it does not work on other machines, then for sure you can object for it by clipping a video that shows it works on your machine.

First and Last Name : Emre Türksoy
Student ID : 2230356030
Link of the Demo Video : https://youtu.be/izB2dJV2_Uw